

Robotics, Automation, and Systems Control Lab (RASCL)

Prof. Dr.-Ing. Mike Barth

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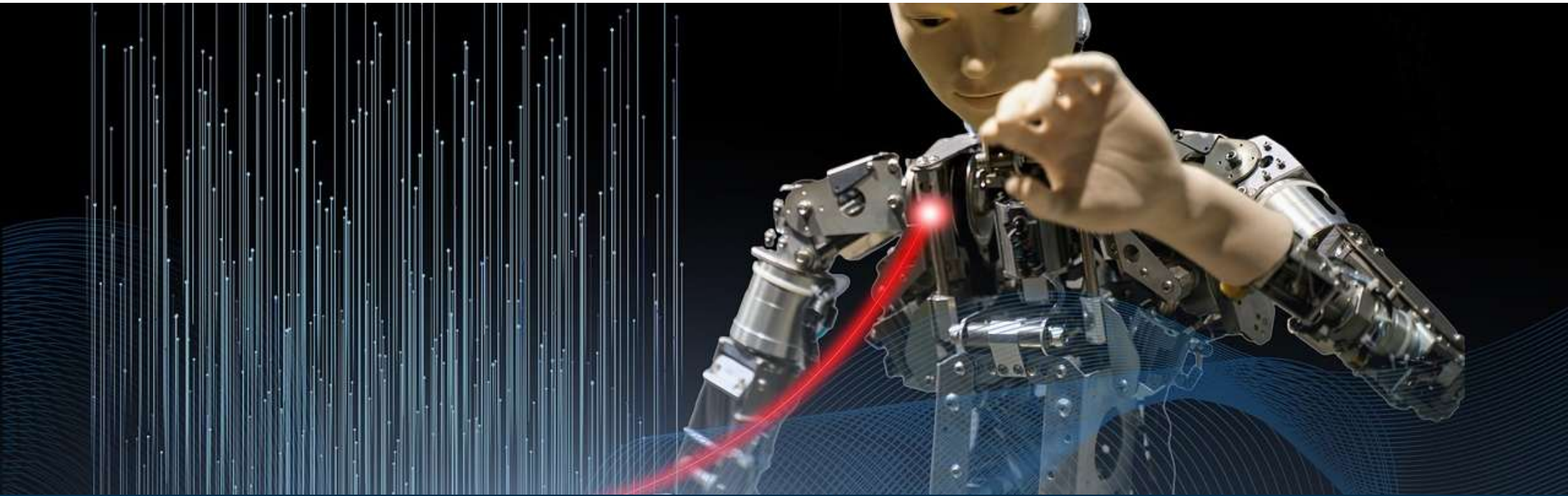
Eric Wagemann

Linus Witucki

Pirmin Breisacher

Agenda 21.04. – Intro Lecture

1. Presentation IRS
2. Presentation Lecturers
3. RASCL – General Introduction
 - a) Robot Demo
 - b) Work Packages
4. Semester Administration
 - a) Semester Schedule
 - b) Grading & Exam Registration
 - c) Further Orga Topics
5. Group Assignment
6. Introduction Work Package 1



Presentation IRS

Chapter
1

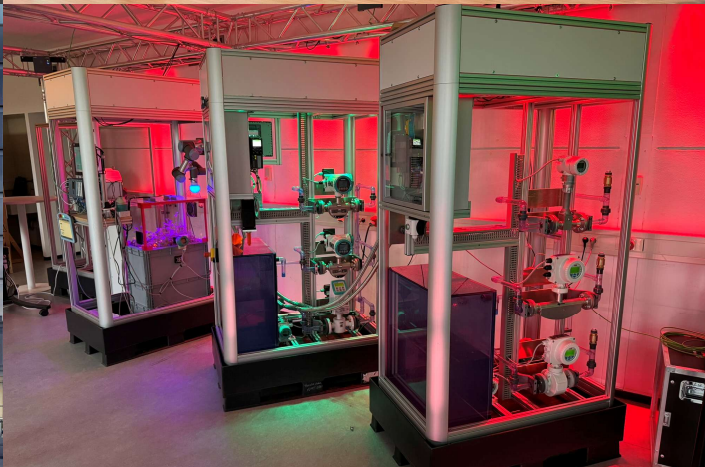
|||||IRS



IRS

Institut für Regelungs-und Steuerungssysteme





IRS at a glance

Networked Multi-Energy Systems

Institute of Control Systems

21.04.2026

Networked Multi-Energy Systems

Background and Vision

Background

- Energy transition requires integration of renewable sources & tapping efficiency potentials – while maintaining secure operation
- Growing number of generators and multi-domain coupling overwhelm classical centralized control approaches
- Operation must remain stable despite uncertainty and structural changes

Vision

- **Heterogeneous subsystems** integrate, coordinate and stabilize autonomously
- **Cross-sector coupling** with robust operation even during volatile feed-in or outages
- **Automated** identification, control and optimization – without complete system knowledge



Networked Multi-Energy Systems

Our Research



We develop and test unified physics-, data- and learning-based methods for the secure operation of networked multi-energy systems

→ Focus on overall system and networking

Electricity

Heat

Gas

Hydrogen

Water

Modular and scalable methods
“Plug and Play”

Unified, integrated
system and control solutions



Stabilization



Coordination



Optimization

Networked Multi-Energy Systems

Core Competences

Secure and efficient operation of heat systems (districts, heat networks) and electricity/heat systems

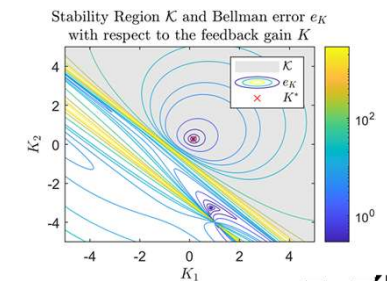
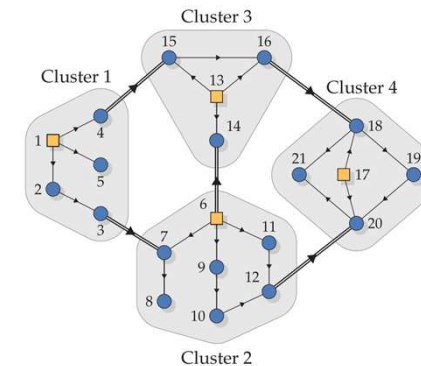
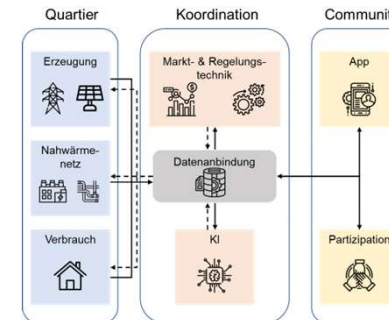
- Stabilization and optimal coordination
- Automation “from planning to optimized online operation”

Decentralized stabilization, coordination and optimization for AC, DC, AC/DC power grids

- Stabilization and optimal coordination (voltage, frequency and power control)
- Unified concepts for “Grid Codes 2.0”

Data- and learning-based control for energy systems with guaranteed stability/performance properties

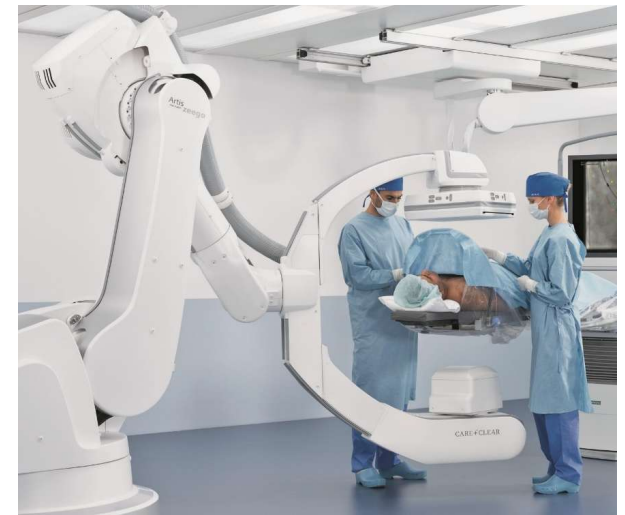
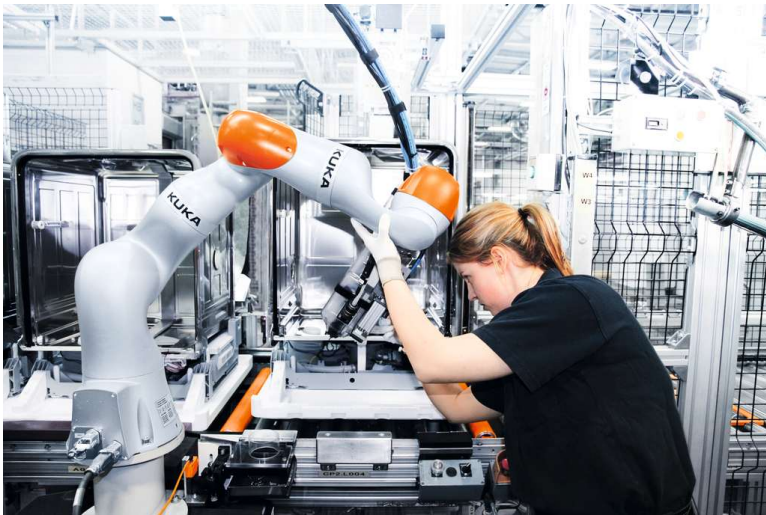
- Geometric learning control for efficient modeling of heat systems
- Policy optimization methods for the Linear Quadratic Regulator



Cooperative Systems

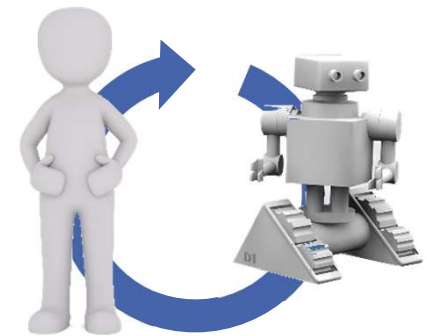


Human-Machine Interactive Systems



Our Vision for Cooperative Systems

Human and machine
mutually **support** and **adapt**, and
cooperate such that they act as **ONE**



Emancipation

Leverage Synergies

Human-Machine Symbiosis

Optimal Interaction

Intermediate Step
towards Full Autonomy

Functional Safety Control

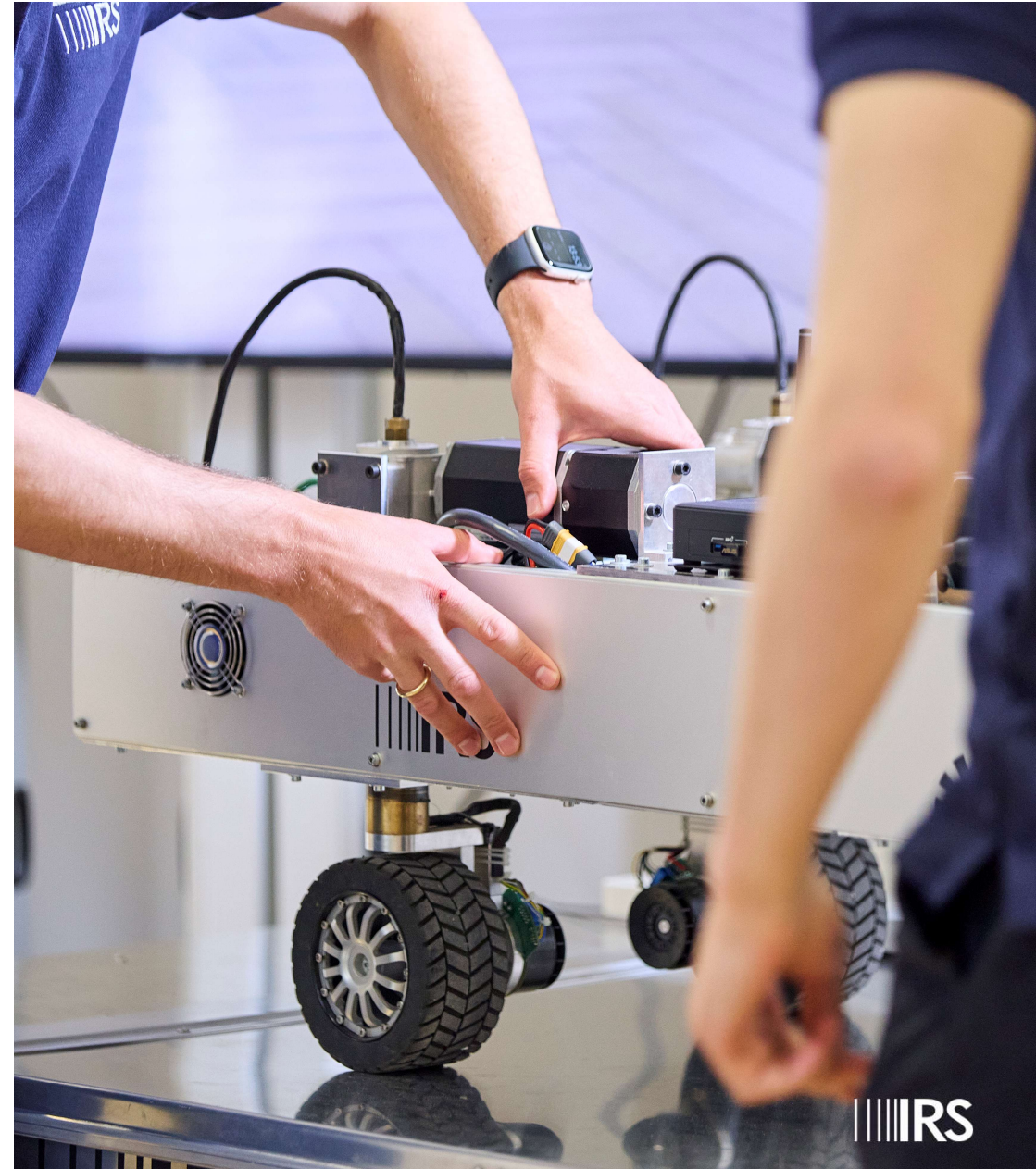
Institute of Control Systems

21.04.2026

Functional Safety Control

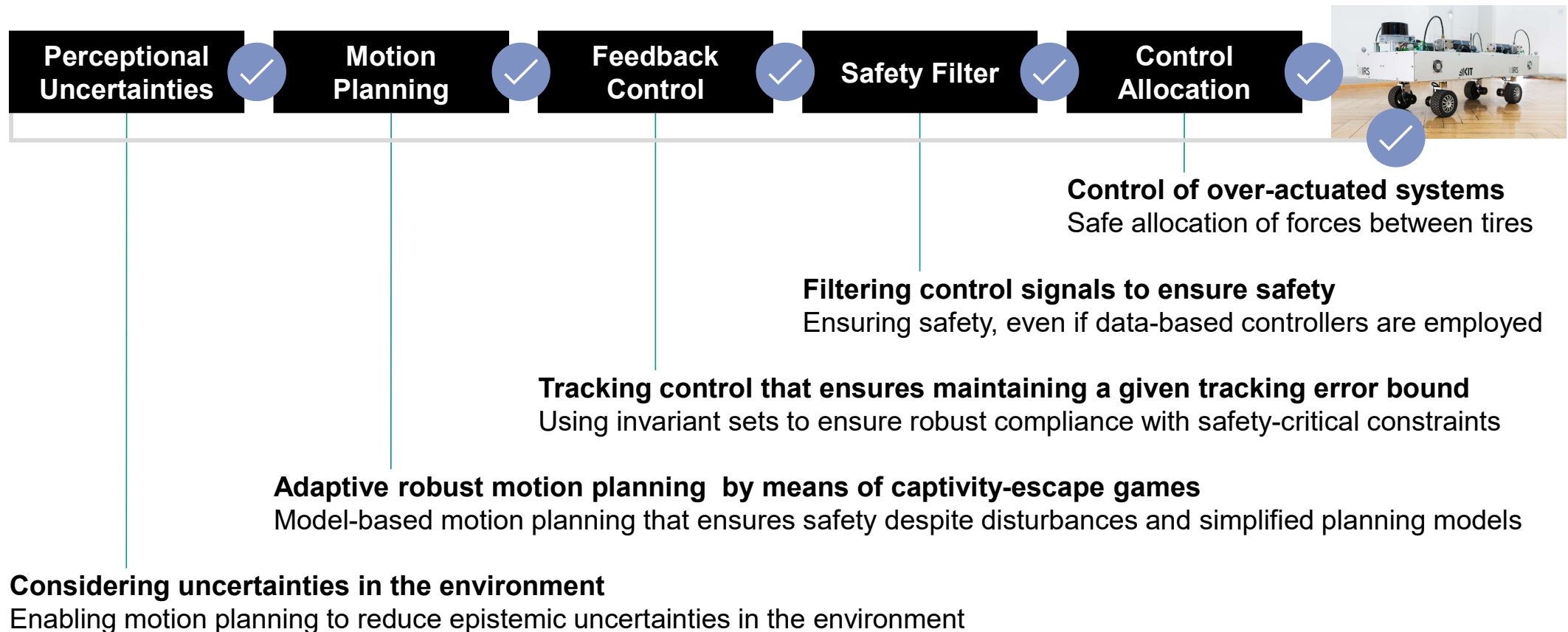
Background and Goal

- Background
 - Obtaining regulatory approval for safety-critical automation systems requires verifying the system's functional safety.
 - As automation systems grow more complex and expand to operate in open environments, relying solely on testing for verification becomes impractical.
 - The current consensus is that formal verification methods are required to ensure the functional safety of a system.
- Goal of our research
 - Our goal is to **formally ensure the functional safety of an automation system by design**.
 - To this end, we research planning and control methods that ensure functional safety **in the presence of disturbances and uncertainties in real-world environments**.



Core Competences

Mobile Robot Example



Information-Driven Engineering of Automation Systems



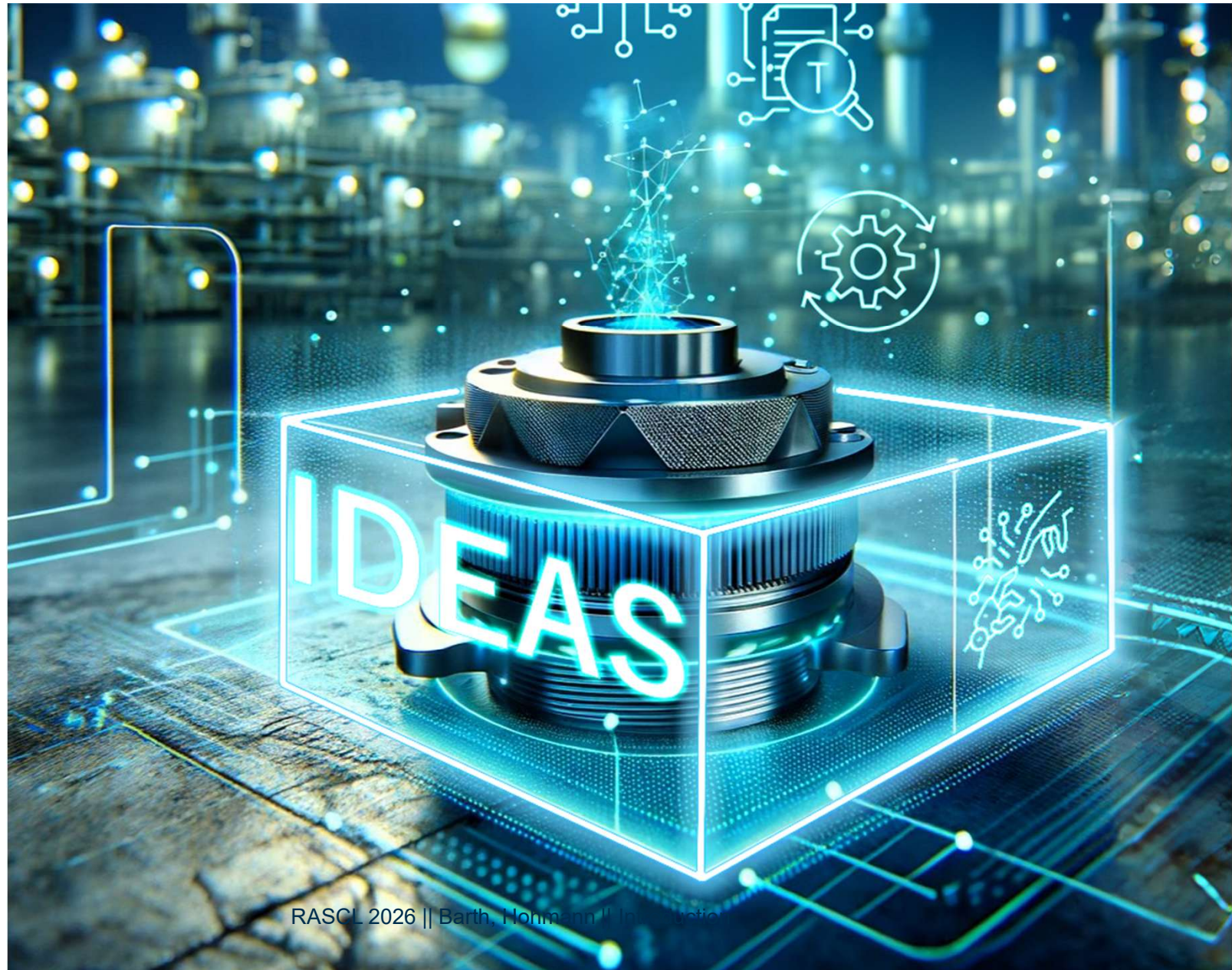
Automation and
assistance
engineering



Model-based
engineering and
operation of
automation systems

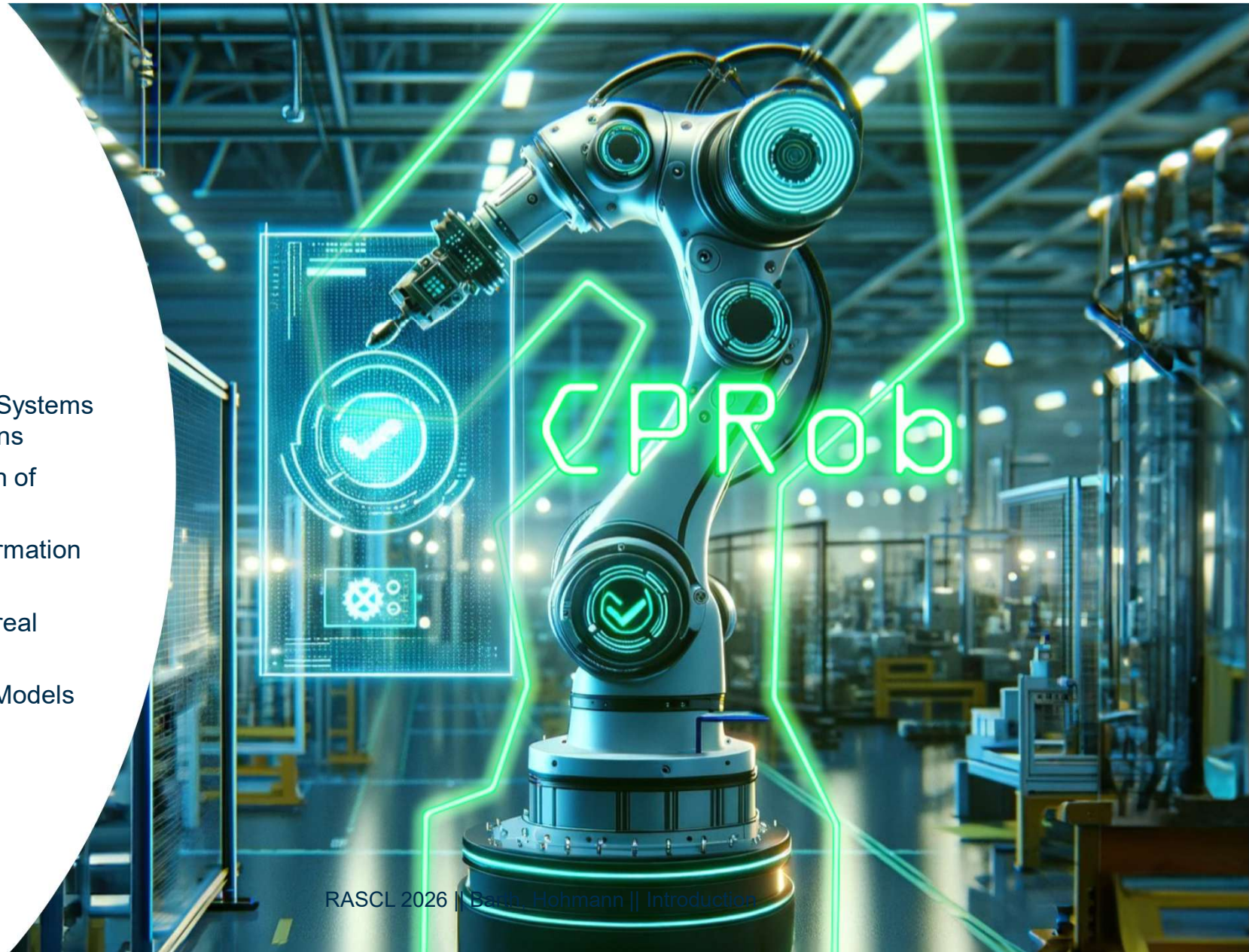


Smart Engineering &
Information Models



Cyber Physical Robotics

- Development of generalized Systems for interacting with digital Twins
- Virtual Reality Implementation of functional Digital Twins
- Generalization based on Information Models
- Coupling of Digital Twin with real Robots
- Development of Information Models for Robots



Cyber-Physical Robotics

Research Artefacts:

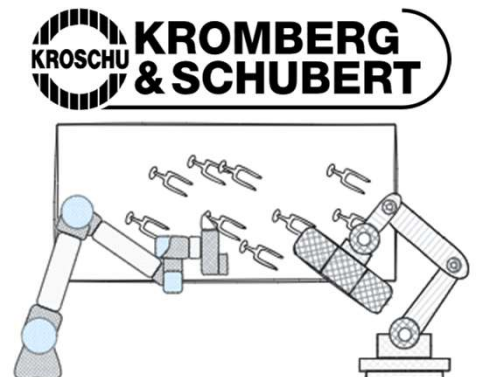
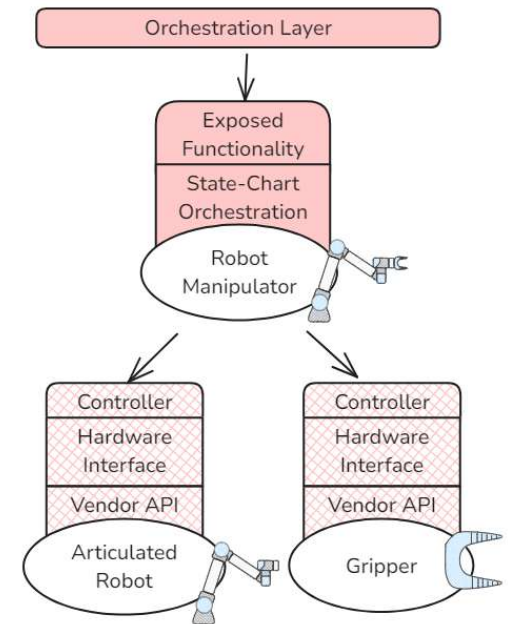
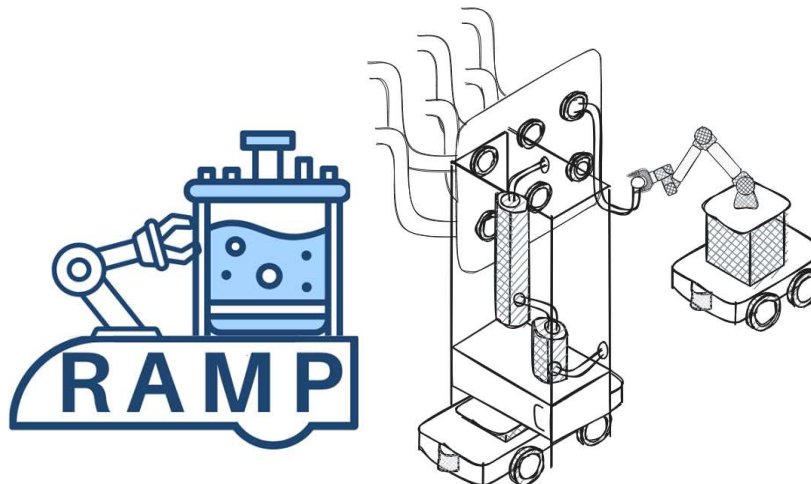
- Software-Architectures and information-driven principles of Industrial Robots
- Orchestration, Choreography and Capability-Composed Robot Systems
- Meta- and Mathematical description of robot system types (Soft-Robots, Continuum Robots etc.)

Methods:

- Industrial Informatics for robot systems, Simulation and Systems Engineering (with focus in industrial robots)

Project Examples:

- Flexible robot orchestration to produce complex wiring systems for the automotive industry
- Robot-Assisted Modular Production (RAMP)
- Soft Robot Architecture



Cyber-Physical Robotics



Cyber-Resilience and System Security



Lifecycle Security
Update Management &
Model-based forecast

Secure Automation
Architectures

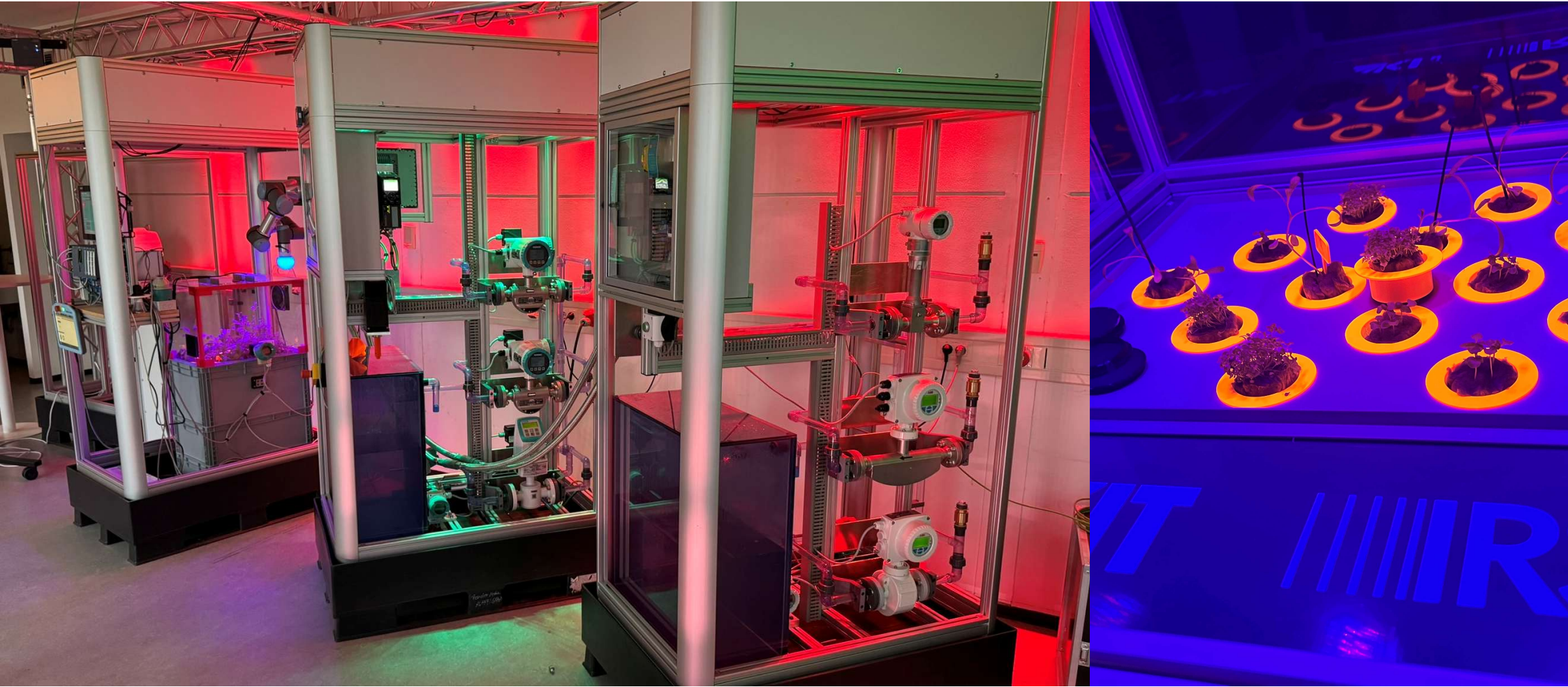


System Security
Methods
Certification
Management

Use of cryptography



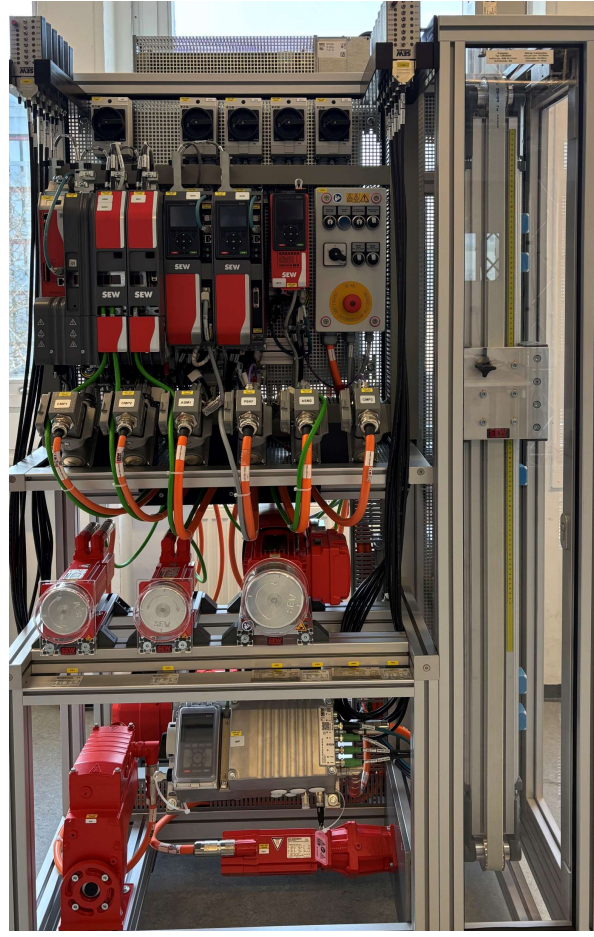
Growing Lab Infrastructure



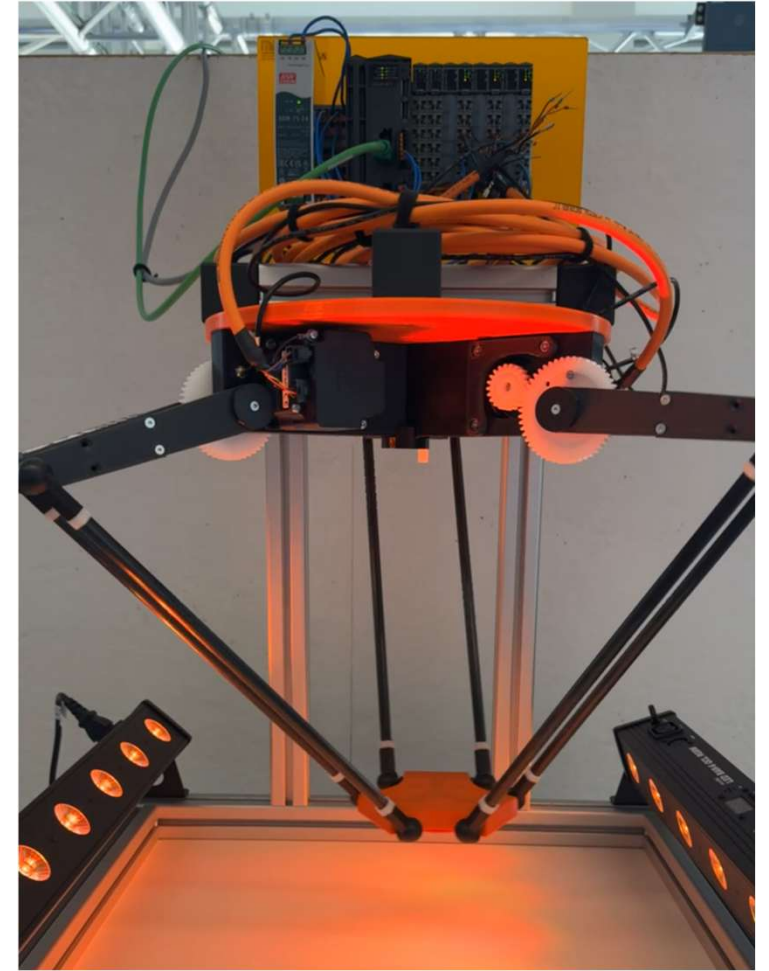
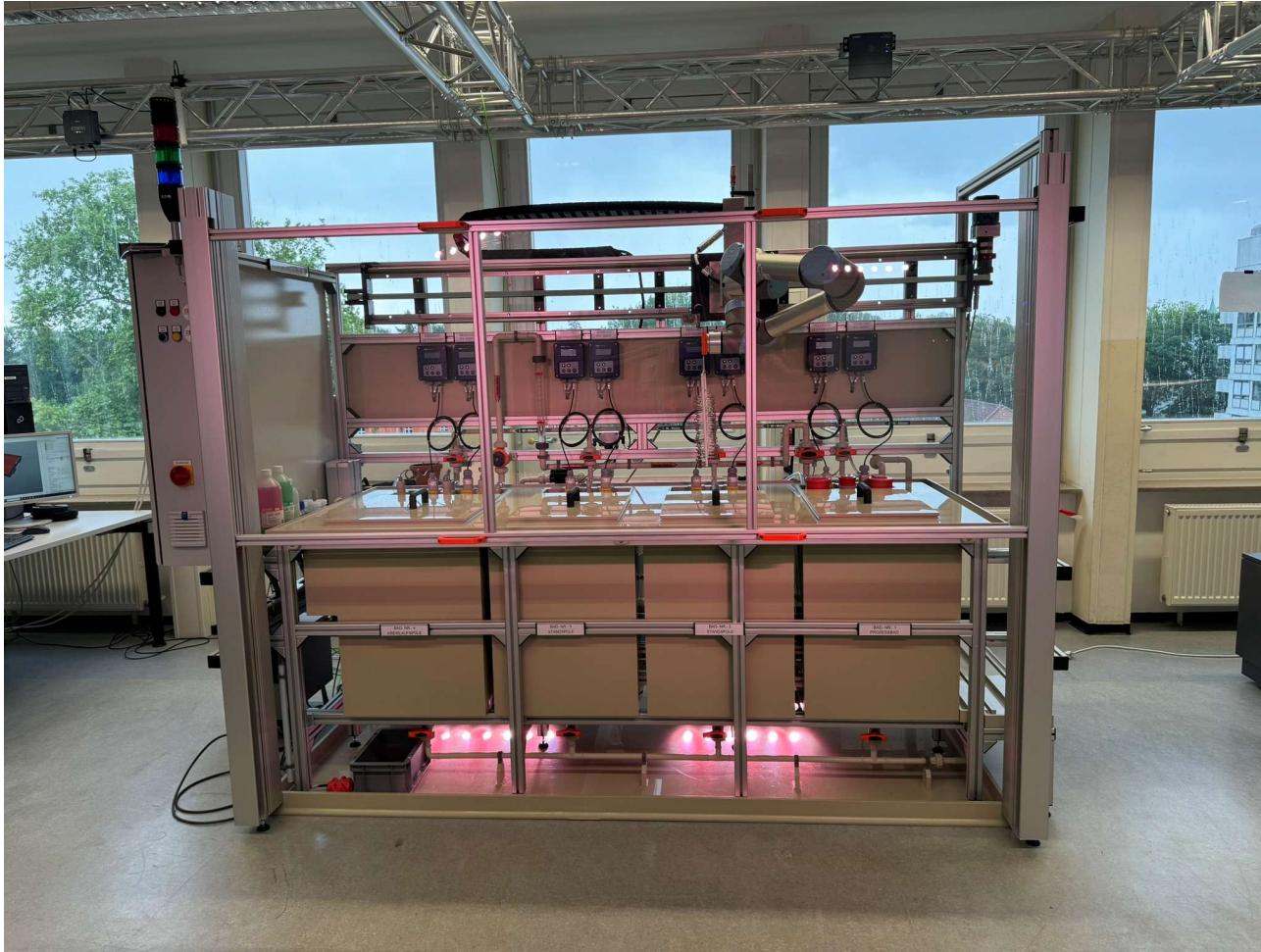
Wachsende Labor-Infrastruktur

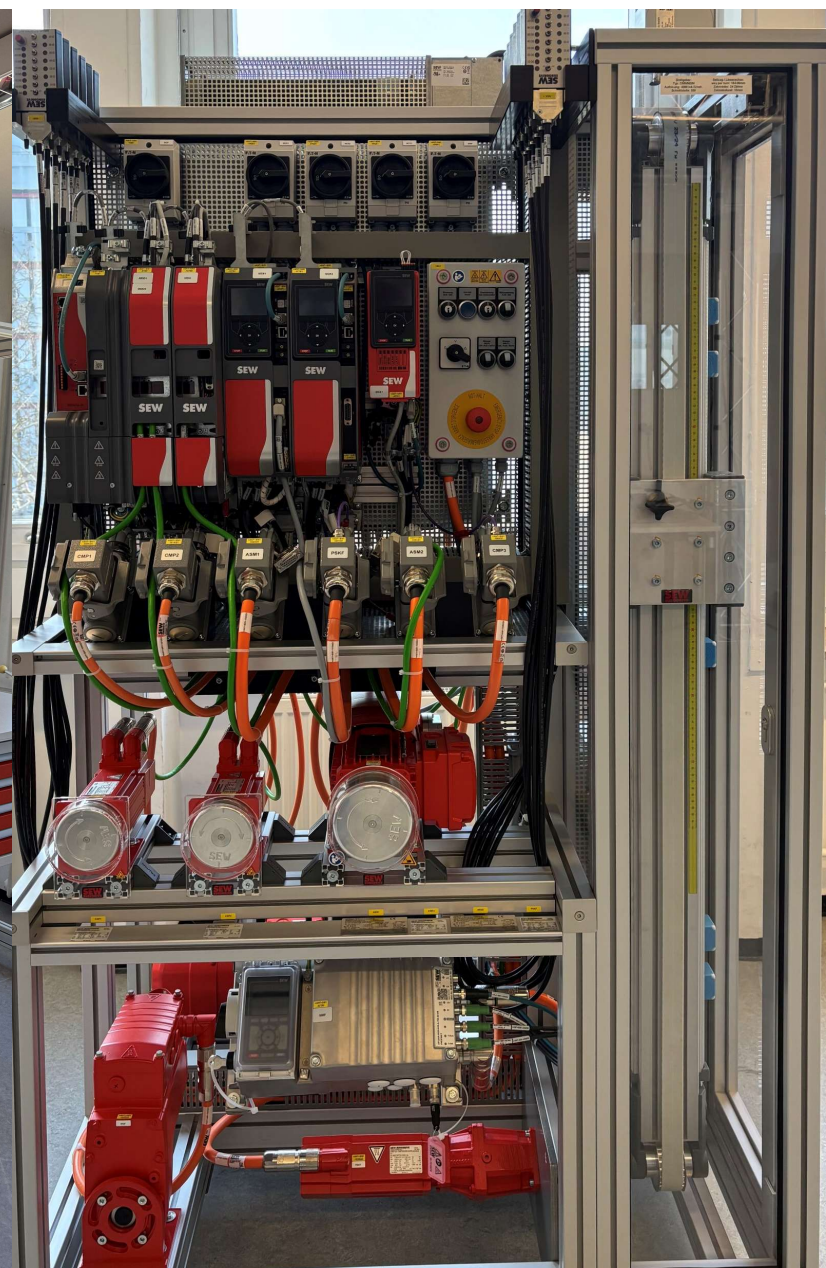


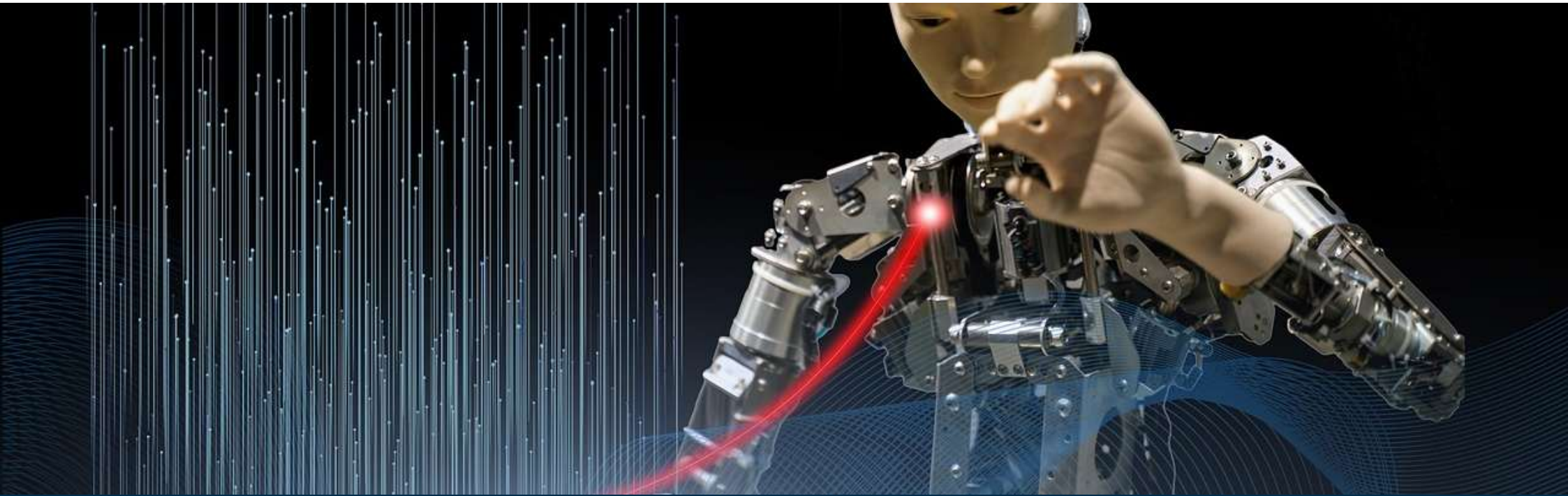
Growing Lab Infrastructure



Growing Lab Infrastructure







Presentation Lecturers

Chapter
2

2 – Presentation Lecturers



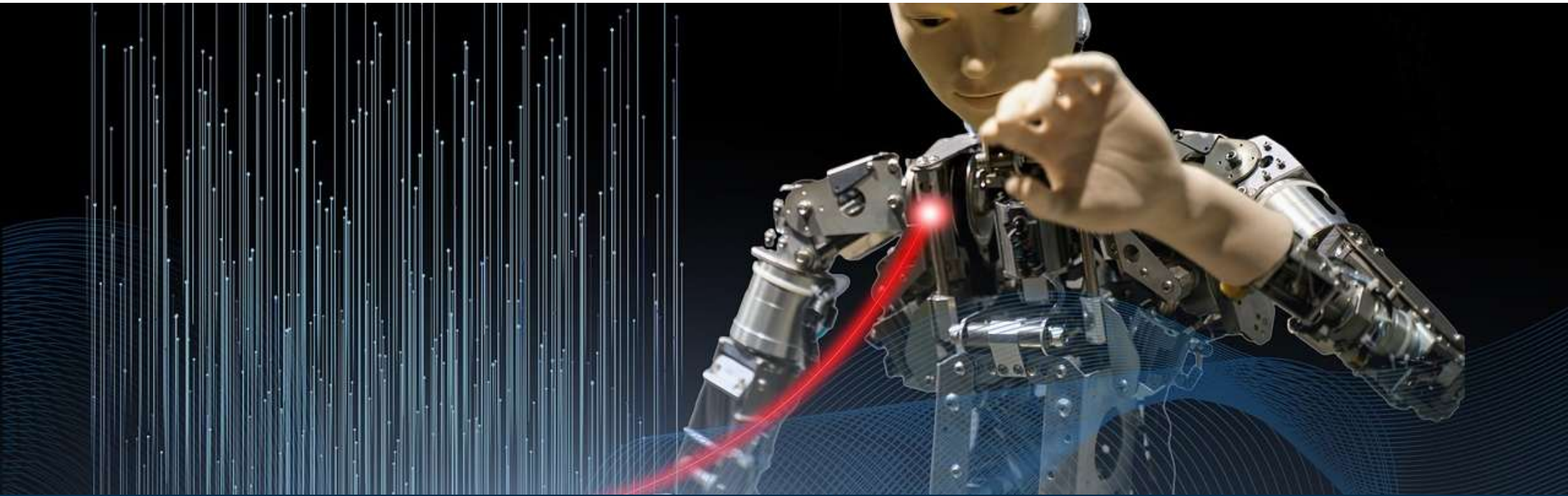
Pirmin Breisacher



Linus Witucki

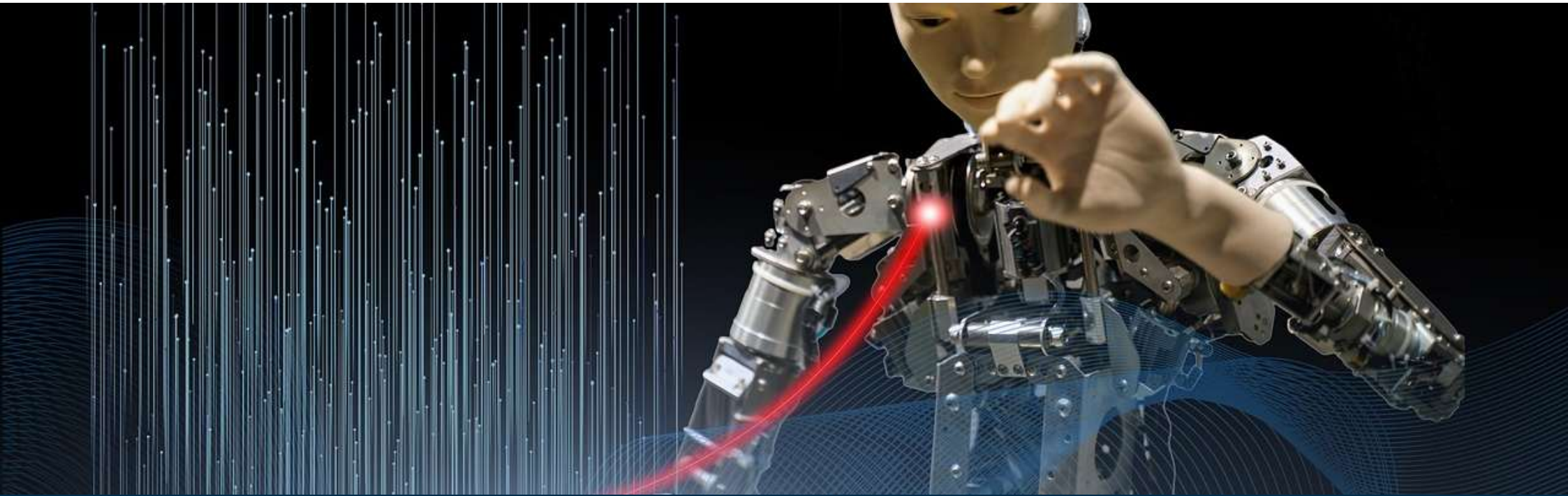


Eric Wagemann



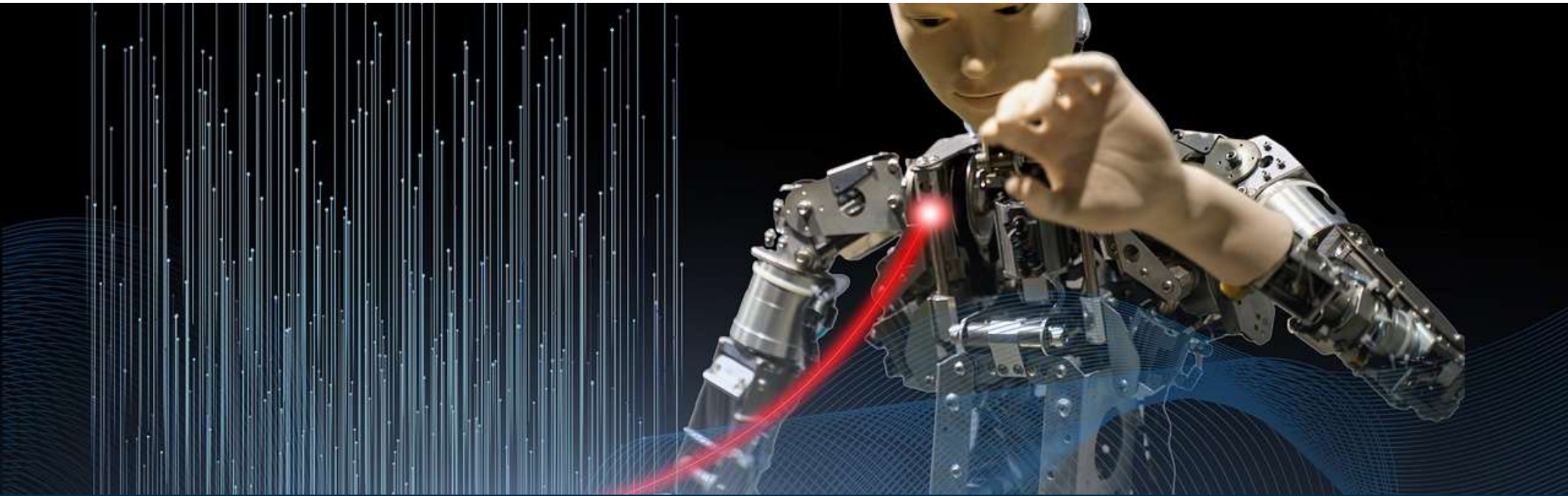
RASCL – General Introduction

Chapter
3



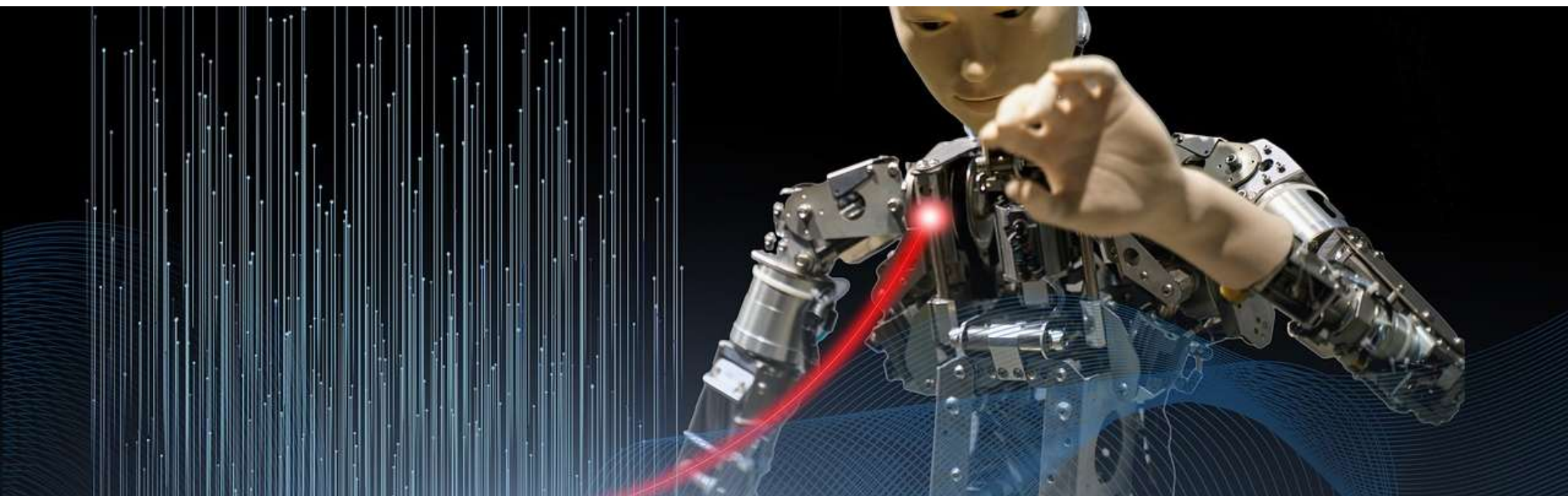
Demo

Chapter
3-1



Work Packages

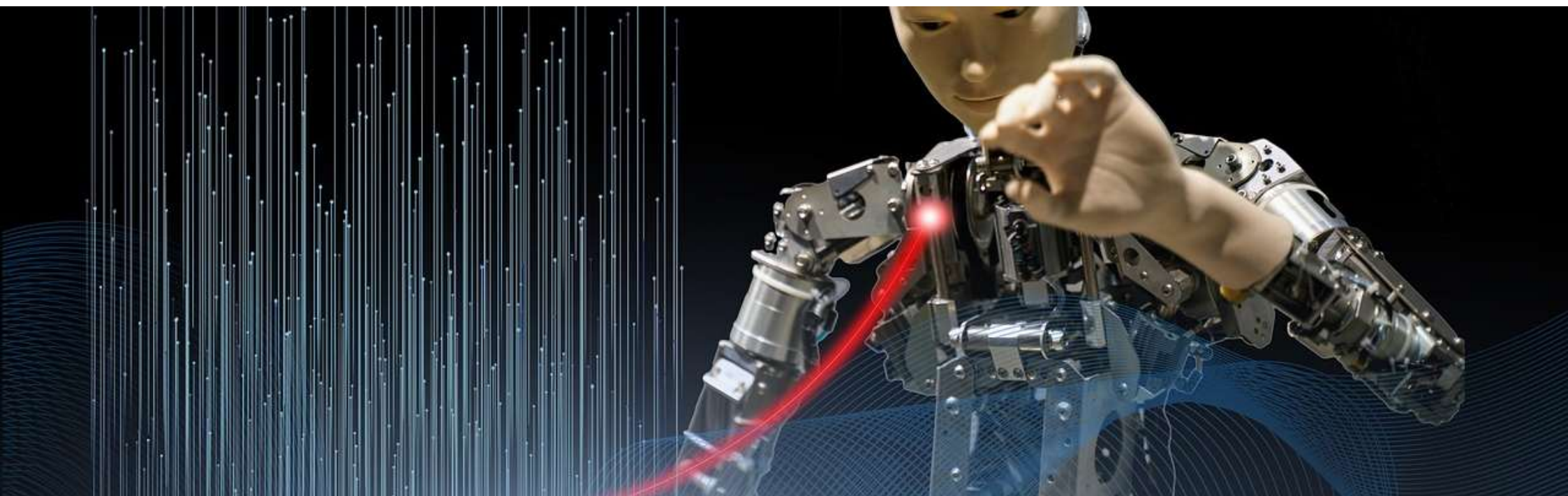
Chapter
3-2



Introduction

Work package 1

Chapter
3-2

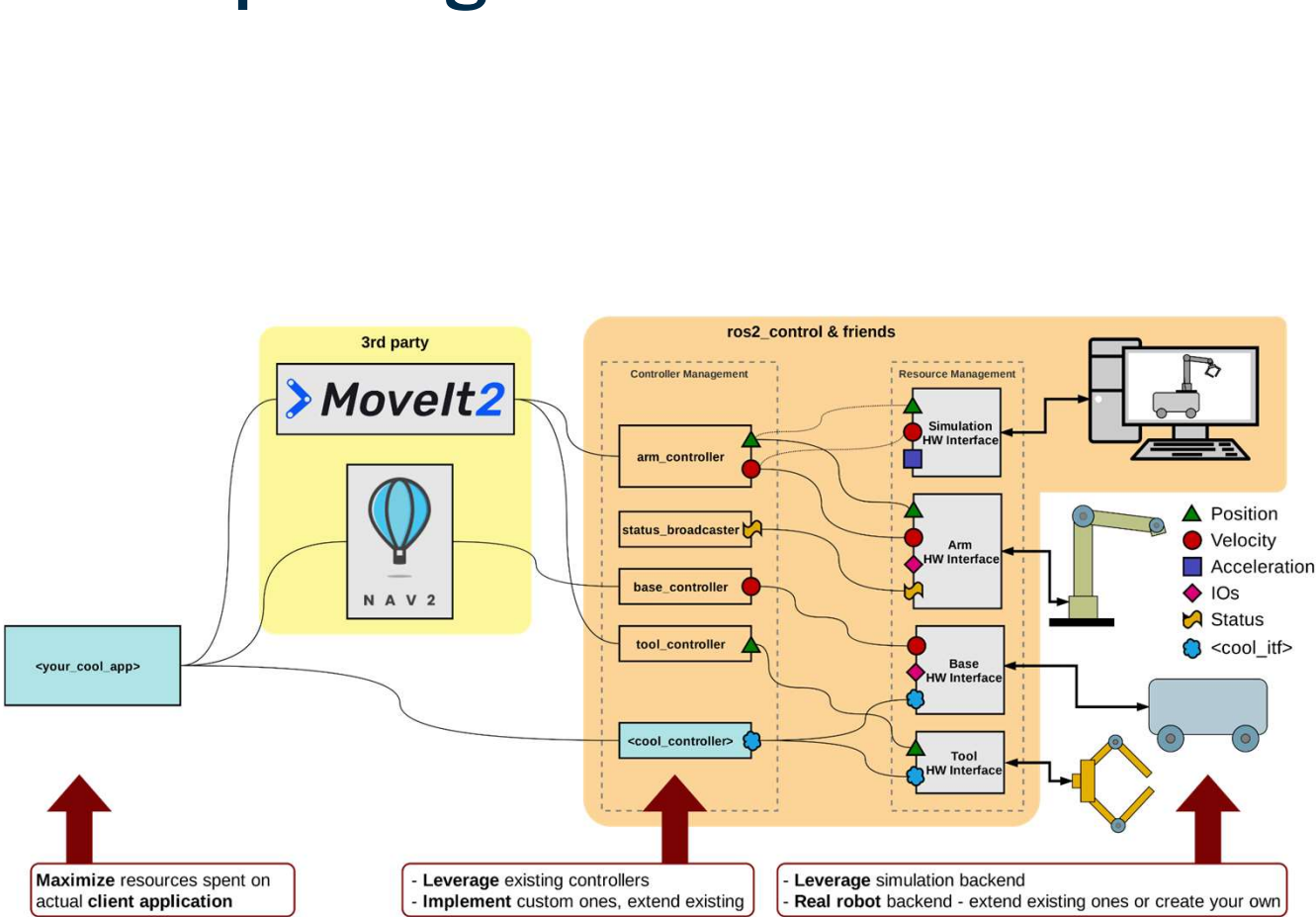


Introduction

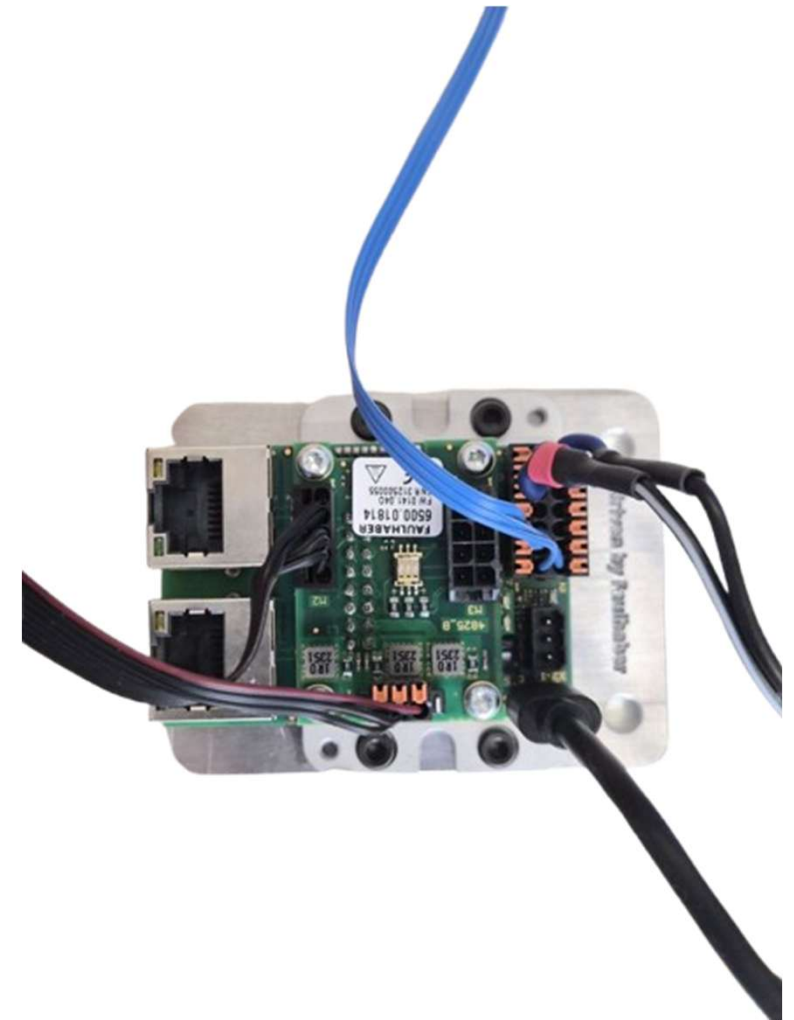
Work package 2

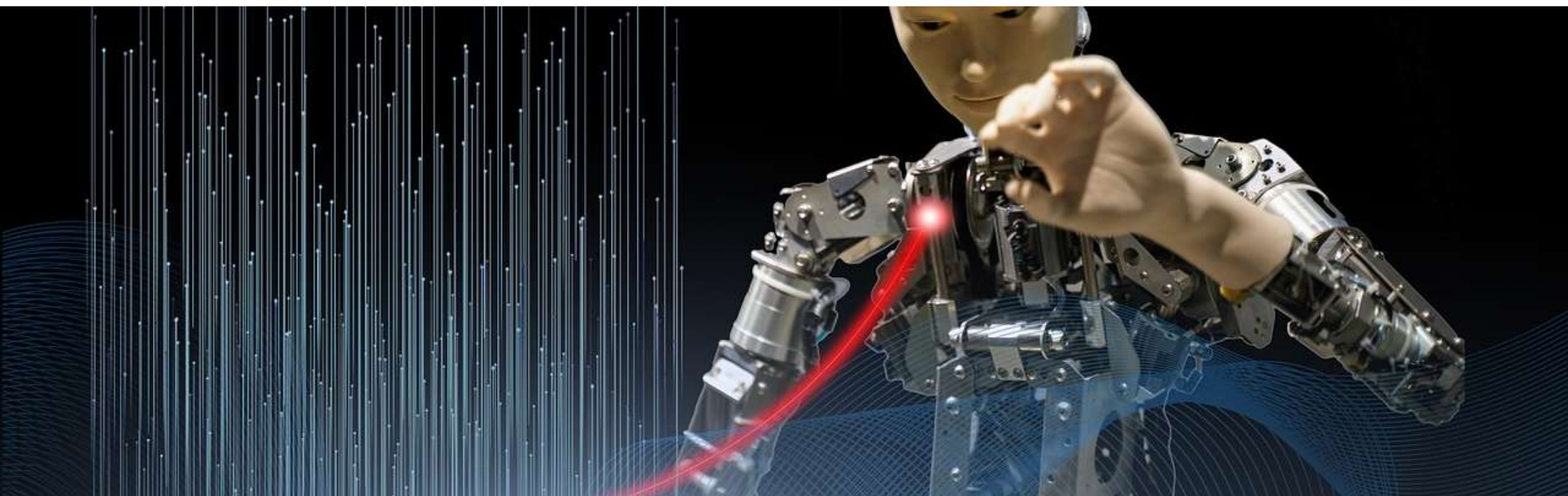
Chapter
3-2

Workpackage 2 – Motion Controller and ROS2 Control



CC-BY: Denis Stogl, Bence Magyar (ros2_control)





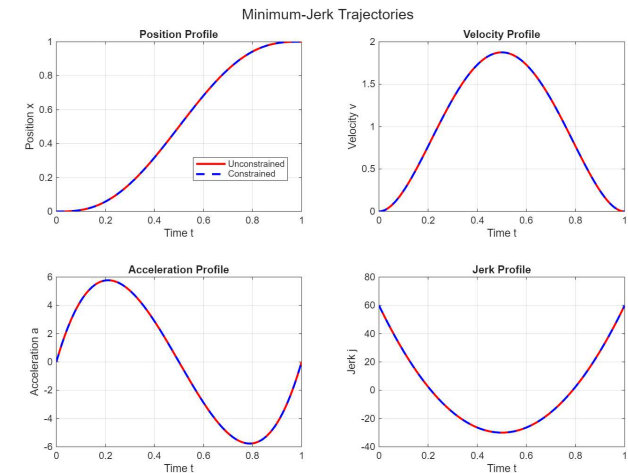
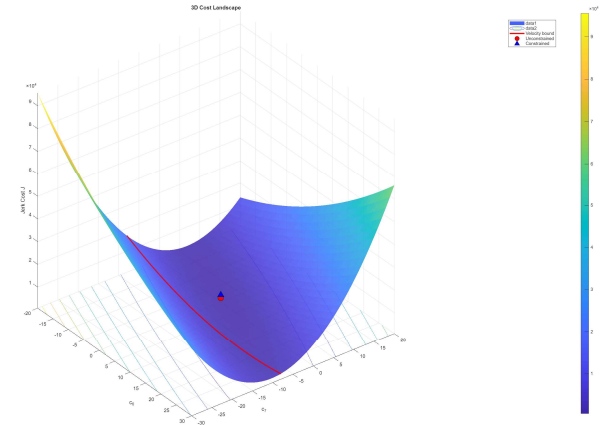
Introduction

Work package 3

Chapter
3-2

WP3 – Automation and Robot Control

- The Task is introduced in full detail in the introductory lecture on the 22.06.
- Task Outline
 - Create an automation concept to control the robot
 - Solve a pick-and-place task
 - Offline planning of minimum-jerk trajectories
 - You may use genAI according to your preference – we care that you understood the underlying concepts of your solution

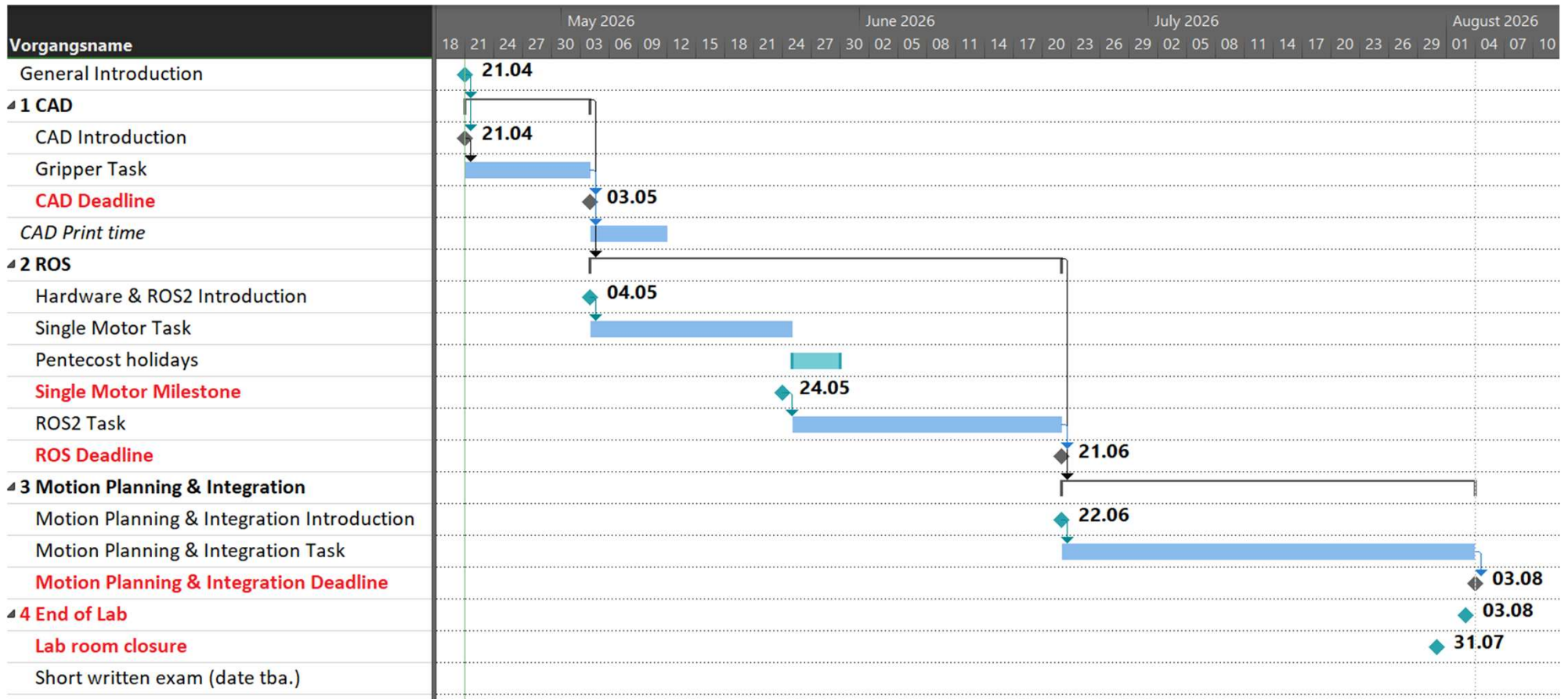




Semester Administration

Chapter
4

Semester Schedule



Grading & Exam Registration

Grading

- WP1: CAD, URDF, Flyer (May 3rd)
- Milestone: Clock Video
- WP2: Software, Robot Video
- WP3: Software, Stacking Video
- Overall: Individual Short Written Exam (approx. 45 min)

Exam Registration

- Registration & Deregistration Deadline: 24.04.2026, 23:59
- Due to necessary restrictions, you'll be on a waiting list. Final registration is done by us.



Further Topics

- **Lab Room**

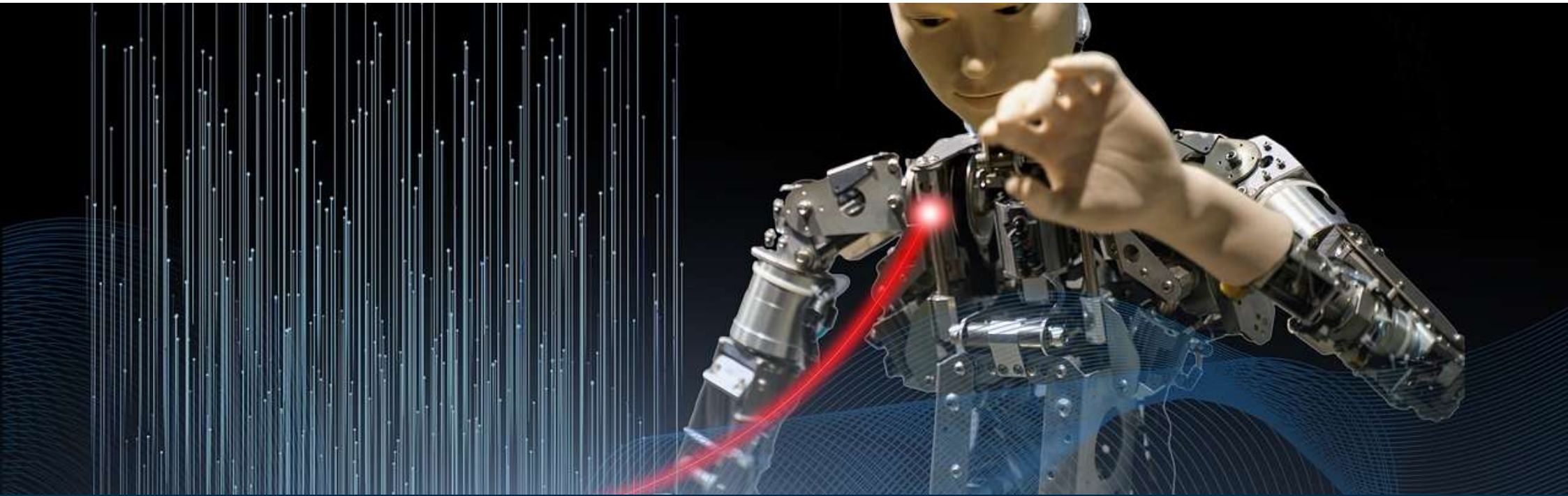
- Building 30.33 R412 (access through the rear door)
- Book your time slot here: <https://plus.campus.kit.edu/signmeup/go/g9UVVv2>
- Closed during Pentecost Holidays and closes finally on 31.07.

- **Intro Lectures**

- 21.04. Intro WP1 (today)
- 04.05. Intro WP2 – Time: 15:45, Venue tba. in Ilias
- 22.06. Intro WP3 – Time: 15:45, Venue tba. in Ilias

- **Questions regarding the tasks**

- Use the consultation hours of the tutors: every Tuesday and Friday 12:00 to 14:00 in the Lab Room (May to July)
- Questions they cannot answer: Ilias forum
- Questions that cannot be answered in Ilias:
 - WP 1: 27.04. 3pm in lab 412
 - WP 2&3: you may make an appointment with the lecturers every Thursday 14:00 to 15:00. Fix it and hand in your dedicated questions at least 24h before to rascl@irs.kit.edu, else you will have to wait for the following Thursday.

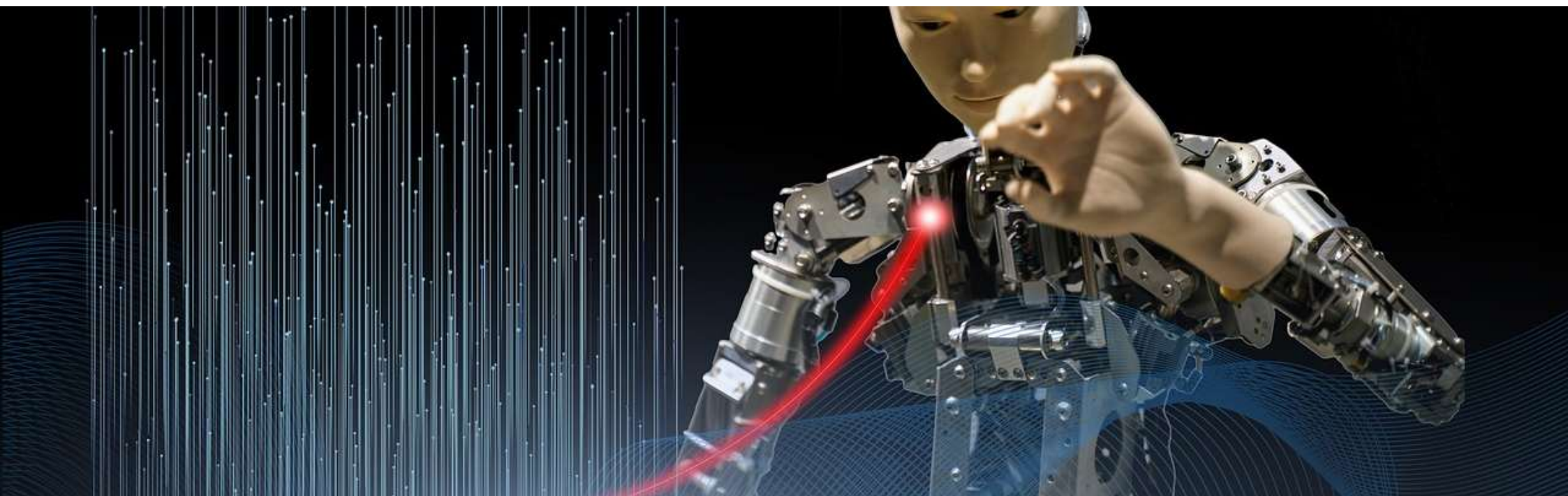


Group assignment

Chapter
5

Group Assignment

- Form teams of three and enter them in Campus+:
<https://plus.campus.kit.edu/signmeup/procedures/6655>
- Any remaining students will be assigned to a group randomly
- The earlier your teams stands, the better as you will have more time in WP1.
- You now have 15 minutes to find teammates



Introduction

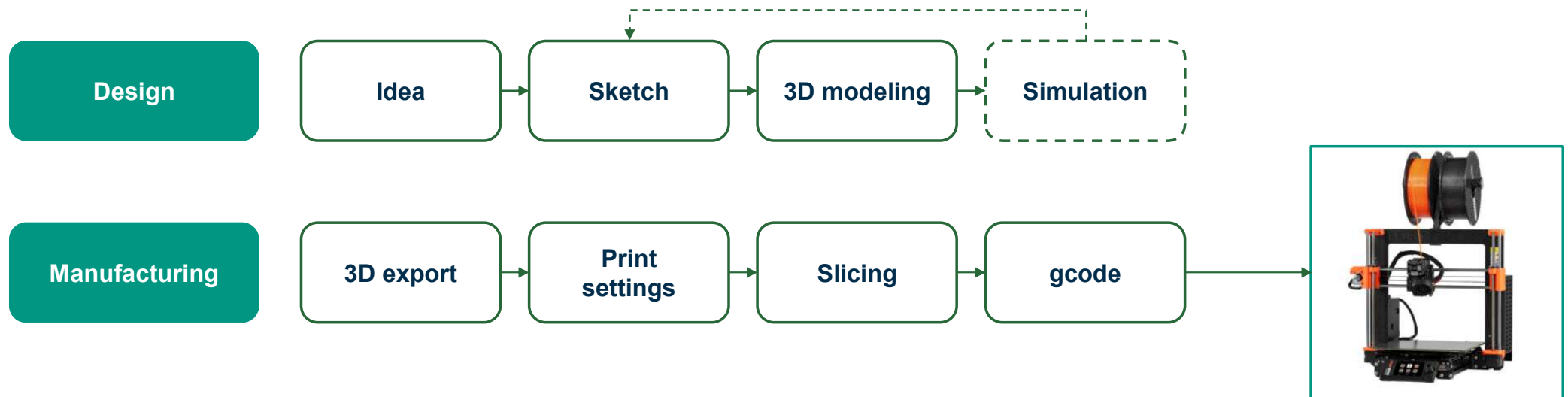
Work package 1

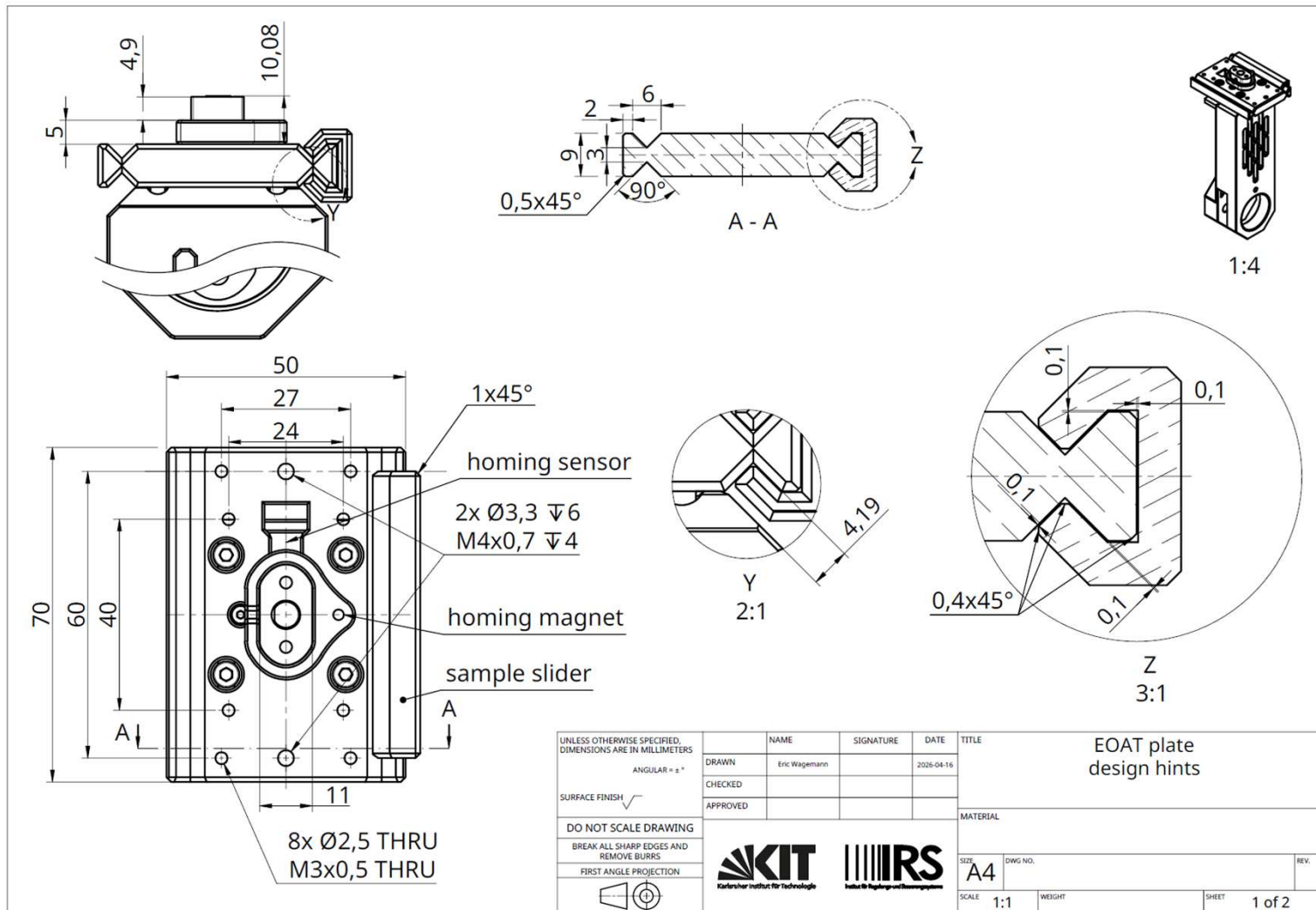
Chapter
6

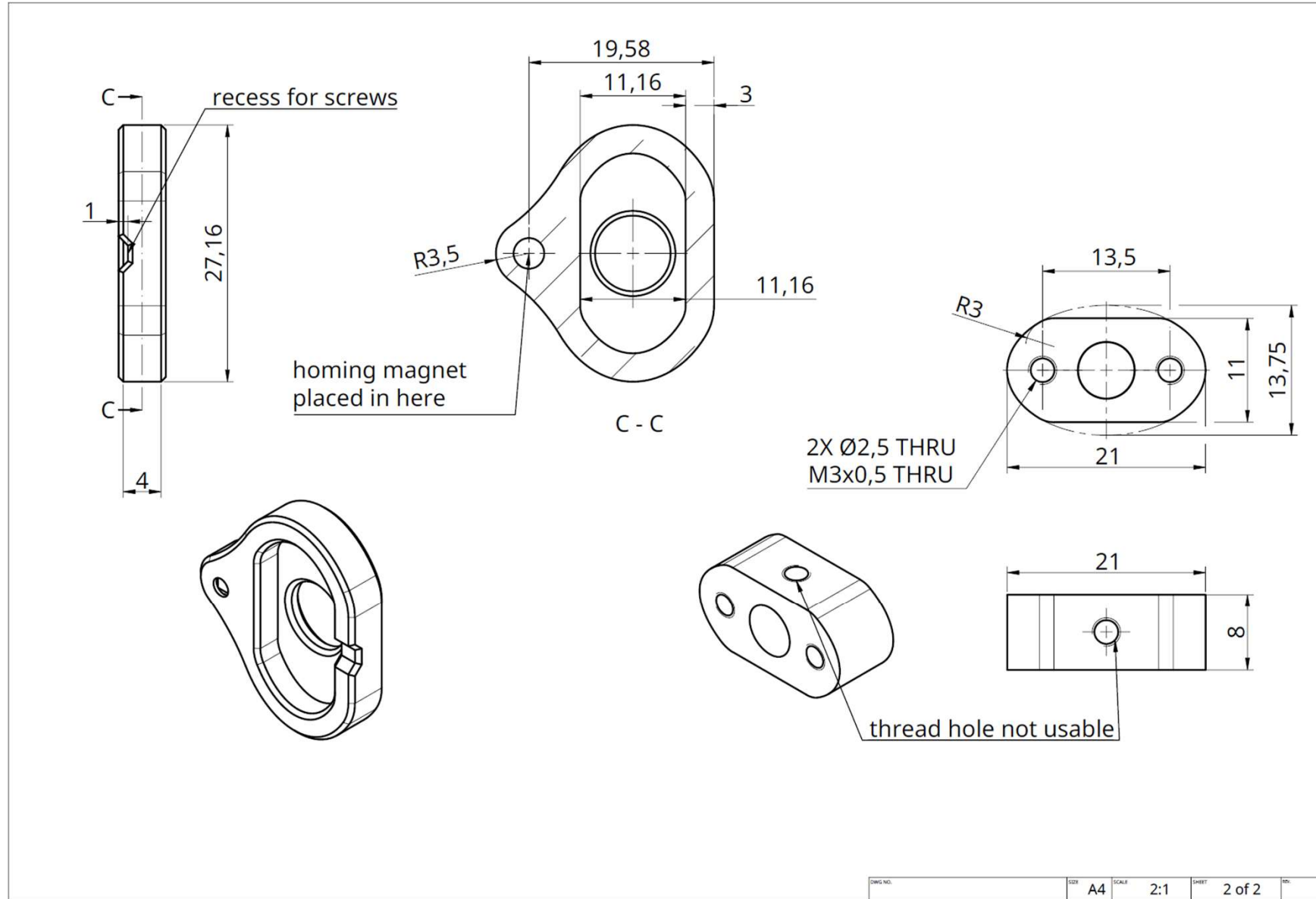
What is a mechatronic system?
What belongs to a mechatronic system?

Brainstorm with your neighbor!

Path to your first 3D printed part





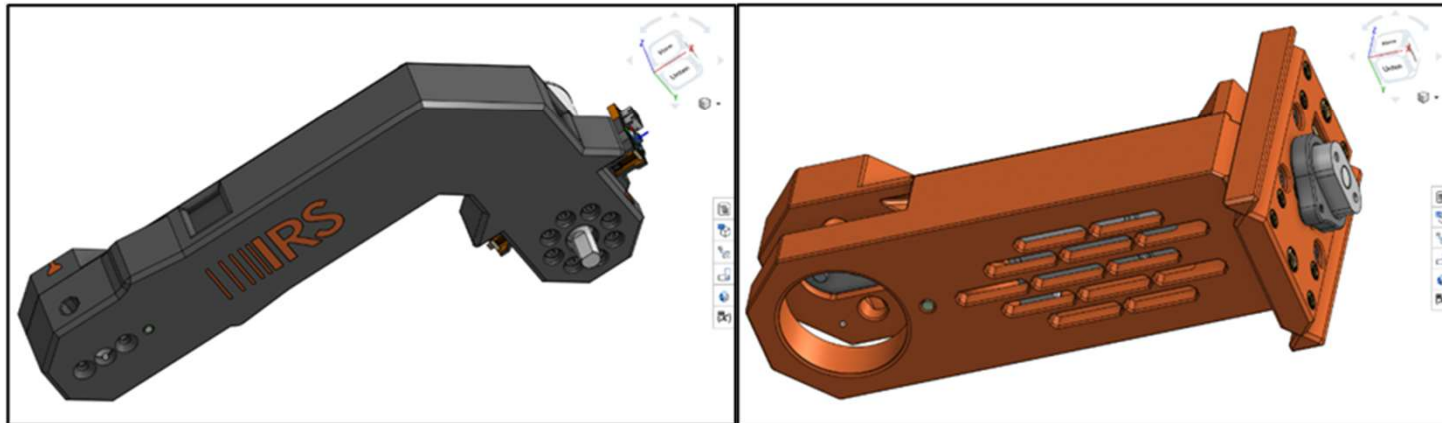
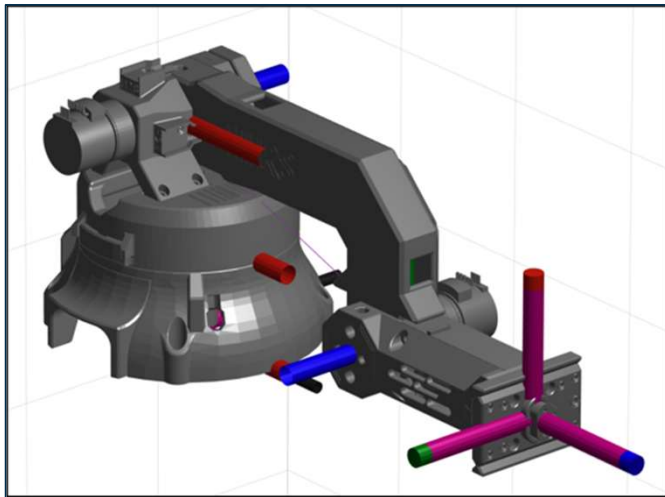
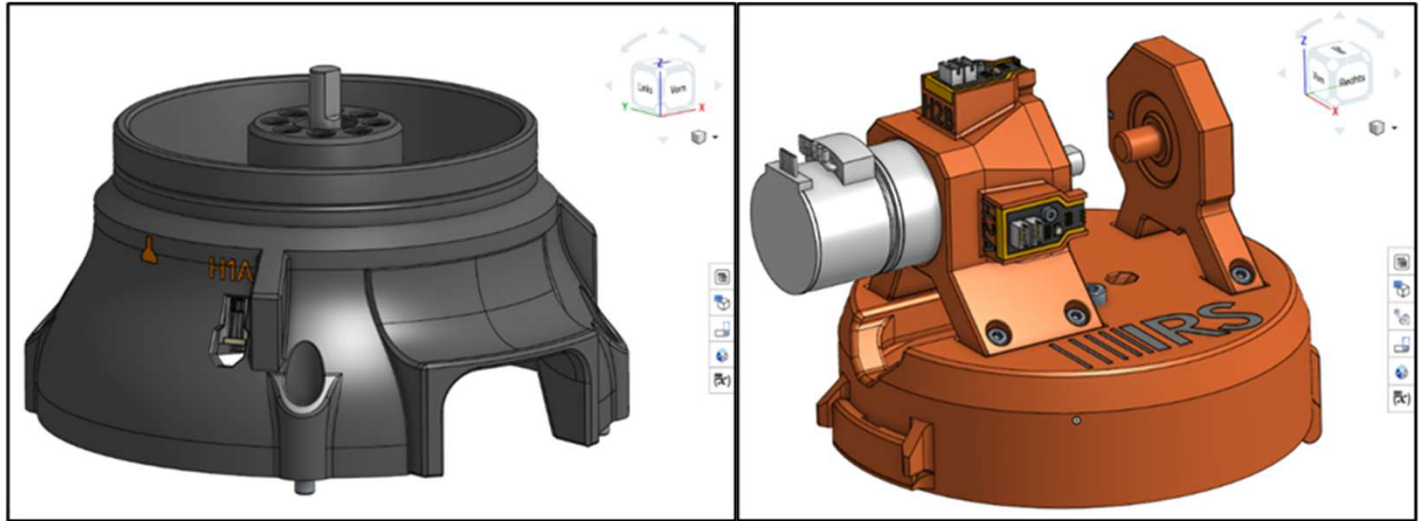


Evaluation guidelines

- Functionality
 - Ability to grip a cube safely
 - Ease of assembly to the robot
- Manufacturing
 - Short printing time
 - Material consumption of printer
 - Use of support materials
 - Amount of additional material
- Creativity/originality of your design
 - What is your key selling point?
- URDF
 - Running URDF file with gripper

URDF

- Unified Robot Description Format
- Information Model
- Common in the ROS environment
- Consists of links and joints
- xml format



What you need to do

CAD

- Design a gripper for a 40mm cube
- Study the drawings regarding the connection dimensions
- Brainstorm in the group, sketch ideas for a gripper
- Familiarize with CAD and onshape (further material is provided via ILIAS course for RASCL)
- Design your part in onshape
- Slice your parts in the Prusa Slicer, save as .3mf
- Make a list of further material (screws, hex nuts, filament)
- Pitch your design with a one-page flyer A4.

URDF

- Familiarize with URDF
- Create the missing part of the URDF file

Office hour on April 27th at 3pm, lab 412 (30.33)

Files are uploaded today or tomorrow.

Submission:

- May 3rd, 2026, 23:59
- via ILIAS group folder
- Details in ILIAS